

Notes: Atkeson and Burstein (AER, 2008)

Pricing-to-Market, Trade Costs, and International Relative Prices

Contents

1	Big picture	2
2	Incremental contribution of the paper	3
3	Data and empirical facts	3
3.1	Price indices and relative prices	3
3.2	Fact 1: smooth terms of trade relative to PPI-based RER	4
3.3	Fact 2: CPI-based RER is about as volatile as PPI-based RER	4
3.4	Evidence at disaggregated levels	5
4	Models and empirical specifications	5
4.1	Key price decomposition	5
4.2	CPI-based RER decomposition	5
4.3	Two-country trade environment	6
4.4	Production and trade costs	6
4.5	Market structure and Cournot competition	6
5	Identification and empirical design	7
5.1	What is identified empirically	7
5.2	Calibration targets	7
5.3	Benchmark and counterfactual models	8
5.4	How to read the quantitative experiment	8
6	Mechanism and intuition	8
6.1	Why variable markups generate incomplete pass-through	8
6.2	Why trade costs are necessary for pricing-to-market	9
6.3	Why large firms drive the aggregate result	9
6.4	Why firm heterogeneity matters	9
7	Tables and figures	9
7.1	Figure 1 and Table 1: Core international relative-price facts	9
7.2	Table 2 and Table 3: Disaggregated evidence and benchmark calibration	10
7.3	Figure 2 and Table 4: Market concentration, markups, and model implications	11
7.4	Figure 3 and Table 5: Firm-level pricing-to-market and sensitivity	12
8	Short summary	13

9	Conclusion	14
9.1	Core conclusion	14
9.2	Economic intuition	14
9.3	Incremental contribution revisited	14
9.4	Implications for trade and macro research	14
9.5	Limitations	15
9.6	Takeaway	15

1 Big picture

- **Question.** Why are international relative prices so volatile, and why do traded-goods prices not obey relative purchasing power parity (PPP) at the aggregate level? The paper asks whether a flexible-price trade model can explain why terms of trade move much less than producer-price-based real exchange rates, while consumer-price-based real exchange rates move almost one-for-one with producer-price-based real exchange rates.
- **Core empirical facts.** The authors organize the paper around two facts for industrialized countries. First, the manufacturing terms of trade are only about one-third to two-thirds as volatile as the manufacturing PPI-based real exchange rate. Second, the CPI-based real exchange rate for goods is roughly as volatile as the manufacturing PPI-based real exchange rate.
- **Economic puzzle.** Under relative PPP, import prices should move one-for-one with source-country producer prices, and export prices should move one-for-one with home producer prices. Then terms of trade should be as volatile as the PPI-based real exchange rate, and consumer-price real exchange rates should be smoothed by trade shares. The data show the opposite pattern.
- **Main mechanism.** Firms engage in *pricing-to-market*: when relative production costs change, a firm changes the relative price it charges abroad versus at home. This pricing response is generated without sticky prices. Prices are fully flexible, but firms have variable markups because they compete imperfectly and their markup depends on market share.
- **Two necessary ingredients.** The model needs both imperfect competition with variable markups and international trade costs. Variable markups generate incomplete pass-through. Trade costs separate the set of competitors faced at home and abroad, allowing markups to move differently across destination markets.
- **Role of market structure.** Firm heterogeneity and within-sector cost dispersion are essential. Large firms account for the aggregate pricing-to-market pattern because they carry large price-index weights. Small firms may price differently, sometimes even in the opposite direction, but they matter little for aggregate indices.
- **Quantitative message.** A calibrated two-country model with nested CES demand, Cournot competition, iceberg trade costs, fixed export costs, and firm productivity dispersion can reproduce the two central relative-price facts. Distribution costs further help the model match the large CPI-based RER movements in the data.
- **Economic takeaway.** Deviations from relative PPP need not be caused only by nominal rigidities. They can arise in a fully flexible-price environment when market power, trade frictions, and heterogeneous firms interact.

2 Incremental contribution of the paper

- **From sticky-price pricing-to-market to flexible-price pricing-to-market.** Much of the open-economy macro literature links pricing-to-market to sticky local-currency prices. This paper shows that pricing-to-market can also arise in a flexible-price equilibrium through strategic markup adjustment.
- **A bridge between international macro prices and heterogeneous-firm trade.** The paper connects real exchange rate and pass-through facts to modern trade models with firm heterogeneity, market shares, export participation, and trade costs.
- **A quantitative model of aggregate deviations from PPP.** The paper does not only document that PPP fails. It builds a calibrated model that matches trade volumes, exporter shares, sectoral concentration, and markups, and then asks whether the same model can generate the observed relative-price movements.
- **Clarifying the separate roles of variable markups and trade costs.** A key contribution is conceptual: variable markups alone generate incomplete pass-through but not pricing-to-market; trade costs alone are not enough without variable markups. Both are needed for aggregate relative PPP deviations.
- **A market-structure contribution.** The paper shows that the distribution of market shares within sectors is central. The pricing behavior of large firms dominates aggregate export and producer price indices, so aggregate pricing-to-market is a weighted result, not a representative-firm result.
- **A technical contribution within nested CES demand.** The authors show that cost dispersion across firms is important for generating pricing-to-market in the empirically relevant direction under a nested CES demand system. Without within-sector cost dispersion, the model can produce pricing-to-market in the wrong direction.
- **A decomposition contribution.** The paper provides clean decompositions linking the terms of trade, producer-price real exchange rates, export-price deviations, import-price deviations, and consumer-price real exchange rates.
- **Policy and empirical relevance.** The framework helps interpret exchange-rate pass-through, trade policy shocks, tariff pass-through, and the connection between market concentration and international price adjustment.

3 Data and empirical facts

3.1 Price indices and relative prices

- The paper studies several price indices: manufacturing producer price indices (PPI), export price indices (EPI), import price indices (IPI), and consumer price indices (CPI) for goods.

- The PPI-based real exchange rate compares a country's manufacturing producer prices with a trade-weighted average of foreign manufacturing producer prices, all expressed in a common currency.
- The terms of trade are measured as the ratio of export prices to import prices.
- The CPI-based real exchange rate compares consumer prices across countries. For the United States in the 1985–2006 period, the paper can focus on goods CPI; for some other countries and longer periods, total CPI is used because goods-only CPI is not always available.

Interpretation: The paper focuses on tradeable-goods price indices rather than services or aggregate CPI alone. This is important because the puzzle is already present for manufactured/traded goods; it is not simply a nontradables story.

3.2 Fact 1: smooth terms of trade relative to PPI-based RER

- In US data from 1985 to 2006, the manufacturing PPI-based real exchange rate moves much more than the manufacturing terms of trade.
- Between 1985 and 1988, the US PPI-based real exchange rate depreciated by roughly 40 percent, while the terms of trade fell by only about 15 percent.
- Across the United States, Japan, Germany, France, Italy, the United Kingdom, and Canada, the terms of trade are typically only one-third to two-thirds as volatile as the PPI-based RER.
- The correlation between terms of trade and the PPI-based RER is positive but far below one.

Interpretation: If aggregate relative PPP held for traded goods, the terms of trade should move approximately one-for-one with relative producer prices. The data show substantial smoothing, implying systematic export and import price adjustment relative to producer prices.

3.3 Fact 2: CPI-based RER is about as volatile as PPI-based RER

- In standard relative PPP logic, consumer prices should average domestic and foreign producer prices according to trade shares, so the CPI-based real exchange rate should be smoother than the PPI-based real exchange rate.
- In the data, CPI-based real exchange rates for goods are roughly as volatile as PPI-based real exchange rates.
- Table 1 shows that the relative volatility of CPI/CPI^* compared with PPI/PPI^* is close to one for most countries and often above one.

Interpretation: The same deviations from relative PPP that smooth the terms of trade also prevent import prices from stabilizing consumer prices. This is why CPI-based RERs remain large and volatile.

3.4 Evidence at disaggregated levels

- The paper documents deviations from relative PPP not only at the aggregate country level, but also in disaggregated import prices by source country and export prices by sector.
- For US imports from Japan, the EU, and Canada, the ratio of US import prices to source-country PPIs is highly volatile and strongly correlated with bilateral PPI-based RERs.
- For US four-digit manufacturing export sectors, the median volatility of PPI/EPI is about 36 percent of the aggregate PPI-based RER, with substantial heterogeneity across sectors.
- For Japanese manufacturing sectors, PPI/EPI deviations are also large and highly correlated with aggregate relative producer prices.

Interpretation: Aggregate PPP deviations are not purely aggregation artifacts. The disaggregated evidence is consistent with firms or sectors changing destination-specific prices relative to costs.

4 Models and empirical specifications

4.1 Key price decomposition

The paper decomposes the PPI-based real exchange rate into terms-of-trade movements and two deviations from relative PPP:

$$\left(\frac{\widehat{PPI}}{\widehat{PPI}^*}\right) = \left(\frac{\widehat{EPI}}{\widehat{IPI}}\right) + \left(\frac{\widehat{PPI}}{\widehat{EPI}}\right) + \left(\frac{\widehat{IPI}}{\widehat{PPI}^*}\right). \quad (1)$$

- PPI/PPI^* is the producer-price-based real exchange rate.
- EPI/IPI is the terms of trade.
- PPI/EPI captures the gap between home producer prices and home export prices.
- IPI/PPI^* captures the gap between import prices and foreign producer prices.

Interpretation: Under relative PPP, the last two terms would be zero, so terms of trade would move one-for-one with the PPI-based RER. In the data, the last two terms are large and positive when the PPI-based RER moves, so terms of trade move much less.

4.2 CPI-based RER decomposition

In a symmetric two-country setting with balanced trade and import share s_M , the relative movement in the CPI-based RER can be expressed as

$$\frac{\widehat{CPI}_1 - \widehat{CPI}_2}{\widehat{PPI}_1 - \widehat{PPI}_2} = 1 - 2s_M \frac{\widehat{EPI}_1 - \widehat{IPI}_1}{\widehat{PPI}_1 - \widehat{PPI}_2}. \quad (2)$$

- A lower import share s_M makes consumer prices more tied to domestic producer prices.
- A smoother terms of trade also makes the CPI-based RER move more closely with the PPI-based RER.

Interpretation: To explain the large CPI-based RER, the model needs both limited import shares and deviations from relative PPP that make terms of trade smooth.

4.3 Two-country trade environment

- There are two symmetric countries producing and trading a continuum of sectors.
- Each sector contains a finite number of firms. Each firm produces a distinct good and draws an idiosyncratic productivity level.
- Final consumption aggregates sectoral outputs through a CES aggregator with elasticity across sectors η .
- Sectoral output aggregates individual goods through a nested CES aggregator with elasticity across goods within a sector ρ .
- The authors impose $1 < \eta < \rho$, so goods within a sector are more substitutable than goods across sectors.

Interpretation: The nested CES structure is what makes market share matter for markups. A firm with a larger sectoral share faces a lower perceived demand elasticity and therefore chooses a higher markup.

4.4 Production and trade costs

- Each firm uses labor with constant returns to scale. Marginal cost depends on the wage, aggregate productivity, and firm-specific productivity.
- Exporting is costly in two ways: an iceberg marginal trade cost $D > 1$ and a fixed labor cost F .
- The iceberg cost creates destination-specific marginal costs. The fixed cost determines which firms participate in export markets.
- Firms with low marginal cost are more likely to export, have larger market shares, and tend to charge higher markups.

Interpretation: The model jointly produces trade volumes, exporter selection, exporter size premia, market concentration, and pricing-to-market.

4.5 Market structure and Cournot competition

Firms compete in quantities. For a firm k selling in country i , the optimal price has the form

$$P_{ijk} = \frac{\varepsilon(s_{ijk})}{\varepsilon(s_{ijk}) - 1} \cdot MC_{ijk}, \quad (3)$$

where s_{ijk} is the firm's market share in the sector and MC_{ijk} is its marginal cost of serving that market. Under quantity competition, the perceived elasticity is a function of market share:

$$\varepsilon(s) = \left[\frac{1}{\rho}(1 - s) + \frac{1}{\eta}s \right]^{-1}. \quad (4)$$

- If a firm has near-zero market share, it perceives the within-sector elasticity ρ and charges a lower markup.
- If a firm has near-one market share, it perceives the lower cross-sector elasticity η and charges a higher markup.
- Since $\rho > \eta$, the markup is increasing in market share.

Interpretation: This is the key pricing mechanism. A cost shock changes a firm's market share; the market-share change changes the markup; the markup change partially offsets the cost shock.

5 Identification and empirical design

5.1 What is identified empirically

- The paper first documents reduced-form empirical facts about relative-price volatility and correlations.
- It then asks whether a calibrated model can reproduce those facts under plausible parameters.
- This is not a causal empirical identification design. It is a quantitative theory exercise disciplined by price, trade, exporter, concentration, and markup moments.

Interpretation: The paper's credibility comes from matching multiple targets and then showing that the mechanism reproduces untargeted or partially targeted price facts.

5.2 Calibration targets

The benchmark model sets production parameters to match three main US manufacturing moments:

- exports plus imports relative to manufacturing gross output: about 16.5 percent;
- fraction of manufacturing plants that export: about 25 percent;
- median sectoral Herfindahl index: about 1,500, interpreted as moderate concentration.

The model also produces plausible implications for exporter intensity, exporter size premia, exporter productivity premia, and average markups.

Interpretation: The calibration is designed to discipline both international trade exposure and domestic market structure. This matters because pricing-to-market is a joint outcome of trade frictions and market concentration.

5.3 Benchmark and counterfactual models

The paper compares three main parameterizations:

- **Benchmark model:** variable markups and trade frictions.
- **Constant-markup model:** sets $\rho = \eta$, so markups do not vary with market share.
- **Frictionless-trade model:** sets $D = 1$ and $F = 0$, so there are no international trade costs.

Interpretation: These comparisons isolate the mechanism. If either variable markups or trade costs are removed, aggregate relative PPP is restored in the relevant sense, and the model fails to match the key relative-price facts.

5.4 How to read the quantitative experiment

- The main quantitative experiment is a fall in country 1 aggregate productivity that generates a 1 percent increase in country 1 aggregate marginal cost relative to country 2.
- The authors compute changes in PPI, EPI, IPI, terms of trade, and CPI-based RER using price-index formulas analogous to statistical practice.
- The central question is how much terms of trade and CPI-based RER move relative to PPI-based RER.

Interpretation: The experiment is a clean laboratory for pass-through and pricing-to-market: relative costs move, firms reoptimize prices, price indices aggregate firm-level responses, and the model's aggregate implications can be compared with the facts.

6 Mechanism and intuition

6.1 Why variable markups generate incomplete pass-through

When a firm's marginal cost rises relative to competitors, it tends to lose market share. Because the markup is increasing in market share, the firm reduces its markup. Its price therefore rises by less than its marginal cost. This is incomplete pass-through.

Economic message: Incomplete pass-through is not assumed. It is generated endogenously by the market-share dependence of markups.

6.2 Why trade costs are necessary for pricing-to-market

Variable markups alone are not enough. Without trade costs, a firm faces the same competitors when selling at home and abroad. Its market shares and markups are then identical across markets, so its domestic and export prices move together. There is incomplete pass-through, but no destination-specific pricing.

Economic message: Pricing-to-market requires segmented competitive environments. Trade costs create that segmentation.

6.3 Why large firms drive the aggregate result

The paper shows that pricing-to-market is heterogeneous across firms. Small firms may have weak or negative pricing-to-market. Large exporters, however, are more likely to cut export markups relative to home markups when home costs rise. Because aggregate price indices are sales-weighted, large firms dominate the producer and export price indices.

Economic message: The aggregate price response is not the response of a typical firm. It is the response of high-market-share firms that receive large weights in price indices.

6.4 Why firm heterogeneity matters

In a version of the model with no productivity heterogeneity and universal exporting, the model generates pricing-to-market in the wrong direction. Productivity dispersion creates large firms with high market shares and strong strategic markup responses, allowing the aggregate model to match the data.

Economic message: Market concentration is not a side detail. It determines the sign and magnitude of aggregate pricing-to-market.

7 Tables and figures

7.1 Figure 1 and Table 1: Core international relative-price facts

Read:

- Figure 1 plots the US manufacturing PPI-based real exchange rate, the manufacturing terms of trade, and the goods CPI-based real exchange rate between 1985 and 2006.
- The PPI-based RER and CPI-based RER move strongly together, while the terms of trade move much less.
- Table 1 generalizes the visual fact across countries and filtering methods.
- In Panel A, for yearly differences, the terms-of-trade volatility relative to the PPI-based RER is 0.38 for the United States over 1985–2006, while the CPI-based RER volatility is 1.08.

- Across the broader country sample, terms-of-trade relative volatility is usually between about 0.37 and 0.66, while CPI-based RER relative volatility is often near one or above one.
- Panel B shows similar patterns with HP-filtered quarterly data.

Economic message: These are the two motivating facts of the paper. The terms of trade are too smooth relative to producer prices, while consumer-price real exchange rates are too volatile to be explained by simple relative PPP logic.

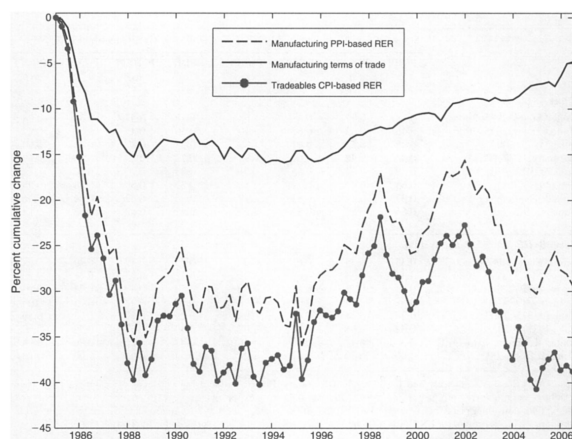


FIGURE 1. US TERMS OF TRADE AND TRADE-WEIGHTED REAL EXCHANGE RATES

(a) Figure 1: US terms of trade and real exchange rates

TABLE 1—VOLATILITY AND CORRELATION OF MANUFACTURING INTERNATIONAL RELATIVE PRICES

	Standard deviation relative to PPI/PPI'				Correlation with PPI/PPI'			
	EPI/IPI (1)	PPI/EPI (2)	IPI/PPPI' (3)	CPI/CPI' (4)	EPI/IPI (5)	PPI/EPI (6)	IPI/PPPI' (7)	CPI/CPI' (8)
<i>Panel A: Yearly differences</i>								
United States, 1985–2006	0.38	0.27	0.68	1.08	0.71	0.57	0.84	0.98
United States, 1975–2006	0.46	0.32	0.67	1.10	0.66	0.45	0.82	0.97
Japan, 1975–2006	0.45	0.53	0.42	1.08	0.53	0.87	0.72	0.99
Germany, 1975–2003	0.58	0.38	0.69	0.95	0.73	0.24	0.70	0.97
France, 1975–2003	0.39	0.64	0.66	0.91	0.23	0.70	0.70	0.81
Italy, 1981–2003	0.65	0.69	0.72	1.11	0.31	0.59	0.54	0.98
United Kingdom, 1975–2003	0.37	0.44	0.63	1.12	0.50	0.62	0.86	0.96
Canada, 1975–2003	0.66	0.50	0.57	1.27	0.58	0.63	0.53	0.85
<i>Panel B: HP-filtered quarterly data</i>								
United States, 1985–2006	0.35	0.35	0.61	1.04	0.76	0.70	0.79	0.97
United States, 1975–2006	0.46	0.34	0.60	1.02	0.72	0.55	0.81	0.96
Japan, 1975–2006	0.42	0.51	0.44	1.09	0.53	0.89	0.74	0.99
Germany, 1975–2003	0.58	0.35	0.64	0.96	0.76	0.23	0.75	0.97
France, 1975–2003	0.36	0.58	0.68	0.95	0.29	0.67	0.74	0.86
Italy, 1981–2003	0.62	0.66	0.70	1.07	0.39	0.53	0.59	0.99
United Kingdom, 1975–2003	0.39	0.48	0.56	1.18	0.53	0.65	0.86	0.96
Canada, 1975–2003	0.60	0.49	0.57	1.24	0.54	0.65	0.63	0.86

Note: All prices are quarterly and measured in US dollars. PPI: manufacturing producer price index; PPI': trade-weighted manufacturing producer price index; EPI: manufacturing export price index; IPI: manufacturing import price index; CPI: consumer price index (goods for US 1985–2006, goods + services for the rest); CPI': trade-weighted consumer price index (goods for US 1985–2006 trading partners, and goods + services for the rest).

Sources: United States: EPI and IPI from Source OECD (1975–1985) and BLS (1986–2006). CPI and PPI from BLS; Japan: EPI, IPI, and PPI from Bank of Japan, CPI from Source OECD; United Kingdom: EPI, IPI, and PPI from UK National Statistics, CPI from Source OECD; Germany, France, Italy, and Canada: EPI, IPI, and CPI from Source OECD. Choice of trade partners is subject to data availability: US, Japan, Germany, France, Italy, UK, Canada, Australia, Austria, Belgium, Denmark, Finland, Greece, Ireland, South Korea, Mexico, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, Singapore, Indonesia, Malaysia, Philippines, Thailand, and India. All

(b) Table 1: Relative volatility and correlations

7.2 Table 2 and Table 3: Disaggregated evidence and benchmark calibration

Read:

- Table 2 documents that relative PPP deviations appear in disaggregated data.
- Panel A shows US import prices by source country. IPI/PPI for Japan, the EU, and Canada is highly correlated with bilateral PPI-based RERs.
- Panel B shows US export-price deviations by four-digit SIC sector. The median sectoral volatility of PPI/EPI relative to the aggregate PPI-based RER is 0.36, but there is wide heterogeneity.
- Panel C shows Japanese export-price deviations by major manufacturing industry, with several correlations above 0.70 and some above 0.90.
- Table 3 reports the benchmark calibration and two comparison models.
- The benchmark model uses $K = 20$, productivity dispersion $\theta = 0.385$, $\eta = 1.01$, $\rho = 10$, iceberg cost $D = 1.45$, and fixed export cost equal to 0.08 percent of the labor force.
- The benchmark matches exports/manufacturing gross output of about 16.6 percent, exporter share of about 24.8 percent, and median Herfindahl of about 1,502.
- It also produces a weighted average markup of about 28.9 percent.

Economic message: Table 2 supports the empirical relevance of pricing-to-market below the aggregate level. Table 3 shows that the model is calibrated to match both trade exposure and market structure, which are exactly the ingredients needed for pricing-to-market.

TABLE 2—VOLATILITY AND CORRELATION OF DISAGGREGATED INTERNATIONAL RELATIVE PRICES

Panel A: US import prices by source country <i>i</i>		
Manufacturing industries HP filtered quarterly, 1991–2006 (IPI source <i>i</i> /PPI <i>i</i>) and (PPI US/PPI <i>i</i>)		
	Ratio of standard deviations	Correlations
Japan	0.85	0.97
EU	0.78	0.98
Canada	0.83	0.85
Panel B: US export prices by SIC industry <i>j</i>		
39 four-digit manufacturing SIC industries, exports to all destinations HP filtered quarterly, 1980–1992, subject to data availability by sector <i>j</i> (PPI US <i>j</i> /EPI US <i>j</i>) and (aggregate PPI US/aggregate PPI* trade weighted)		
Statistics across sectors	Ratio of standard deviations	Correlations
Median	0.36	0.19
Mean	0.49	0.16
Maximum	2.23	0.68
Minimum	0.18	−0.58
Standard deviation	0.40	0.33
Panel C: Japanese export prices by major manufacturing industry <i>j</i>		
HP filtered, quarterly, 1975–2006 (PPI Jap <i>j</i> /EPI Jap <i>j</i>) and (aggregate PPI Jap/aggregate PPI* trade weighted)		
Statistics across sectors	Ratio of standard deviations	Correlations
Textiles	0.59	0.73
Chemicals and related products	1.02	0.73
Metals and related products	0.97	0.74
General machinery and equipment	0.38	0.92
Electrical machinery and equipment	0.63	0.92
Transportation equipment	0.66	0.90

(a) Table 2: Disaggregated relative-price deviations

TABLE 3—PARAMETER VALUES AND SYMMETRIC EQUILIBRIUM IMPLICATIONS

	Benchmark model (1)	Constant markup (2)	Frictionless trade (3)		
Panel A: Parameter values					
<i>K</i>	20	20	20		
θ	0.385	0.385	0.385		
η	1.01	3.5	1.01		
ρ	10.0	3.5	10.0		
<i>D</i>	1.45	1.58	1.00		
Share of labor force in export fixed costs (percent)	0.08	1.33	0.00		
Panel B: Symmetric equilibrium implications					
Exports/manufacturing gross output (percent)	16.6	16.6	50.0	16.5	US Data
Fraction of exporting firms (percent)	24.8	25.6	100.0	25.0	Source
Median exporter's intensity (percent)	11.5	24.3	50.0	< 10	BJ
Average domestic exporter's sales/average nonexporter's sales	14.9	4.8	n.a.	4.8–28	BEJK and EKK
Average exporter's-nonexporter's log value added per worker (percent)	12.5	0.0	n.a.	[15–33]	BEJK
Median Herfindahl index	1,502	740	1,367		
Weighted average markup (percent)	28.9	33.6	26.9		

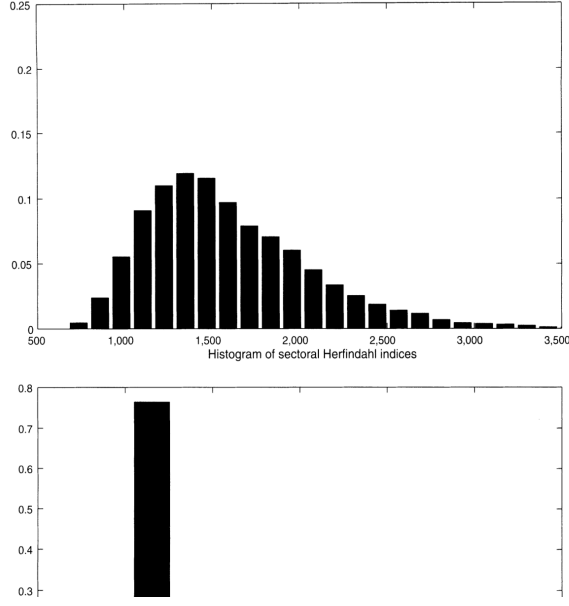
(b) Table 3: Parameters and symmetric equilibrium implications

7.3 Figure 2 and Table 4: Market concentration, markups, and model implications

Read:

- Figure 2 plots the simulated distribution of sectoral Herfindahl indices and firm markups in the benchmark model.
- The Herfindahl distribution is centered around moderately concentrated markets, with the benchmark median close to 1,500.
- The markup distribution is skewed: most firms have relatively modest markups, but some high-market-share firms charge much higher markups.
- Table 4 reports the model's response to a 1 percent increase in country 1 relative marginal costs.
- In the benchmark model, the terms of trade move only 53.4 percent as much as the PPI-based RER.
- The deviations from relative PPP are split almost evenly: PPI/EPI accounts for 23.1 percent and IPI/PPI* accounts for 23.6 percent.
- The CPI-based RER, excluding distribution, moves 82.3 percent as much as the PPI-based RER.
- In the constant-markup model and frictionless-trade model, terms of trade move 100 percent as much as the PPI-based RER, meaning aggregate relative PPP holds.

Economic message: The benchmark model matches the central facts because it has both market-share-dependent markups and trade costs. The counterfactual models show that removing either variable markups or trade frictions destroys the mechanism.



(a) Figure 2: Herfindahl indices and markups

in Table 4 we report on our benchmark model's implications for the relative price movements that are the focus of our study: (a) the movement in the terms of trade, and (b) the movement in the CPI-based RER, both as a percentage of the movement in PPI-based RER. We also include in the table the implications of the constant markup and the frictionless trade versions of our model for these same relative price movements.

TABLE 4—IMPLICATIONS OF MODEL: 1 PERCENT INCREASE IN $(W_1/A_1)/(W_2/A_2)$

	Benchmark model (1)	Constant markup (2)	Frictionless trade (3)
<i>Panel A: Terms of trade and PPI-based RER</i>			
$\widehat{PPI}_1 - \widehat{PPI}_2$ decomposition (percent contribution)			
1 $\widehat{EPI}_1 - \widehat{IPI}_1$	53.4	100.0	100.0
2 $\widehat{PPI}_1 - \widehat{EPI}_1$	23.1	0.0	0.0
3 $\widehat{IPI}_1 - \widehat{PPI}_2$	23.6	0.0	0.0
Change in price indices (percent)			
4 $\widehat{PPI}_1$	0.86	1.00	0.76
5 $\widehat{EPI}_1$	0.69	1.00	0.76

(b) Table 4: Model implications for relative prices

7.4 Figure 3 and Table 5: Firm-level pricing-to-market and sensitivity

Read:

- Figure 3 plots firm-level pricing-to-market against foreign market share for country 1 exporters after a 1 percent increase in country 1 relative marginal costs.
- The vertical axis is the change in the firm's domestic price relative to its export price, scaled by the PPI-based RER movement.
- Small exporters often have zero or negative pricing-to-market; large exporters show positive pricing-to-market.
- The aggregate result is positive because large exporters receive large weights in producer and export price indices.
- Table 5 examines robustness to several model changes: setting fixed export costs to zero, larger cost shocks, increasing the number of firms per sector to 100, restoring Herfindahl concentration, and changing elasticities.
- When all firms export ($F = 0$), the terms-of-trade movement is 55.9 percent rather than 53.4 percent, very close to the benchmark. This shows that fixed export costs and exporter extensive margins are not quantitatively central for small shocks.

- Increasing K to 100 while lowering concentration makes terms of trade move 67.5 percent as much as the PPI-based RER, reducing but not eliminating pricing-to-market.
- Restoring the median Herfindahl to 1,500 with $K = 100$ makes pricing-to-market even stronger: terms of trade move only 46.9 percent as much as the PPI-based RER.
- Reducing the gap between ρ and η weakens the mechanism, but the deviations remain meaningful.

Economic message: The sensitivity analysis identifies market-share dispersion and concentration as the key market-structure force. Export fixed costs matter less than the distribution of firm market shares.

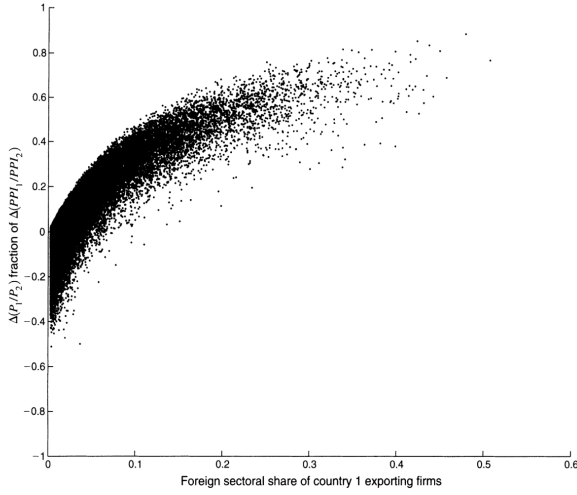


FIGURE 3. FOREIGN MARKET SHARE AND PRICING-TO-MARKET BY COUNTRY 1 EXPORTERS, 1 PERCENT INCREASE IN $(W_1/A_1)/(W_2/A_2)$

(a) Figure 3: Foreign market share and pricing-to-market

	Benchmark model (1)	All firms export $F = 0$ (2)	10 percent change in relative costs (3)	50 percent change in relative costs (4)	$K = 100$ θ at benchmark (5)	$K = 100$ Herfindahl = 1,500 (6)	$\eta = 1.5$ (7)	$\rho = 10$ (8)
<i>Panel A: Parameter values</i>								
K	.20	.20	.20	.20	.100	.100	.20	.20
θ	0.39	0.39	0.39	0.39	0.39	0.57	0.31	0.64
η	1.01	1.01	1.01	1.01	1.01	1.01	1.50	1.01
ρ	10	10	10	10	10	10	10	5
D	1.45	1.46	1.45	1.45	1.34	1.48	1.35	1.85
Share of labor force in export fixed costs	0.08	0.00	0.08	0.08	0.03	0.00	0.11	0.27
<i>Panel B: Symmetric equilibrium implications</i>								
Exports/manufacturing gross output	16.6	16.7	16.6	16.6	16.5	16.6	16.5	16.6
Fraction of exporting firms	24.8	100.0	24.8	24.8	25.0	25.3	24.8	25.2
Median Herfindahl index	1,502	1,493	1,502	1,502	900	1,498	1,513	1,516
Weighted average markup	29	29	29	29	21	29	22	42
<i>Panel C: Change in relative marginal costs (percent)</i>								
$PPI_1 - PPI_2$ decomposition (percent contribution)	1	1	10	50	1	1	1	1
$EPI_1 - IPI_1$	53.4	55.9	53.2	55.1	67.5	46.9	68.9	76.3
$PPI_1 - EPI_1$	23.1	21.8	20.6	16.1	15.7	26.1	15.3	11.7
$IPI_1 - PPI_2$	23.6	22.3	26.2	28.8	16.8	27.0	15.9	12.0
$(\widehat{CPI}_1 - \widehat{CPI}_2)/(\widehat{PPI}_1 - \widehat{PPI}_2)$ (excluding distribution)	82.3	81.4	82.6	86.4	77.7	84.4	77.3	74.7

(b) Table 5: Sensitivity analysis

8 Short summary

- The paper studies why relative PPP fails for traded goods in aggregate price data.
- The two motivating facts are: terms of trade are much smoother than PPI-based real exchange rates, and CPI-based real exchange rates are about as volatile as PPI-based real exchange rates.
- The paper builds a two-country model with nested CES demand, firm heterogeneity, Cournot competition, iceberg trade costs, and fixed export costs.
- Markups are variable because firms with larger market shares face lower demand elasticities.
- When a home cost shock raises marginal costs, firms lose market share and reduce markups, generating incomplete pass-through.
- Trade costs make home and foreign competitive environments different, so markup adjustment differs across domestic and export markets.

- This produces pricing-to-market: firms change export prices differently from domestic prices.
- The benchmark model matches trade volume, exporter share, market concentration, and markups.
- It generates terms-of-trade movements about 53 percent as large as PPI-based RER movements and CPI-based RER movements about 82 percent as large before adding distribution costs.
- With distribution costs, the CPI-based RER becomes about as volatile as the PPI-based RER, matching the data more closely.
- Large firms drive aggregate pricing-to-market because price indices are expenditure-weighted.
- Fixed export costs and extensive-margin export participation are not the main quantitative drivers for small shocks; within-sector market-share dispersion is central.

9 Conclusion

9.1 Core conclusion

Atkeson and Burstein show that a flexible-price model of imperfect competition, variable markups, firm heterogeneity, and trade costs can reproduce major international relative-price facts. The paper’s central conclusion is that aggregate deviations from relative PPP can arise endogenously from firms’ optimal pricing decisions, without assuming sticky prices.

9.2 Economic intuition

The intuition has three steps. First, market power makes pass-through incomplete because firms adjust markups when their market shares change. Second, trade costs make the home and foreign markets different, so the same firm changes its markup differently at home and abroad. Third, heterogeneous firm size means large firms dominate aggregate price indices, so the aggregate pattern reflects the pricing behavior of high-market-share firms.

9.3 Incremental contribution revisited

The paper advances the international macro and trade literatures by showing how modern trade ingredients can explain real exchange rate and pass-through facts. It clarifies that variable markups are necessary for incomplete pass-through, but trade costs are necessary for pricing-to-market. It also shows that market concentration and firm heterogeneity are not just micro details; they shape aggregate price-index behavior.

9.4 Implications for trade and macro research

The paper suggests that empirical work on exchange-rate pass-through, tariff pass-through, and international price adjustment should pay close attention to market structure. Shocks may be transmitted differently in concentrated markets than in competitive markets. Large exporters may

drive aggregate pass-through, while the behavior of average or small exporters may be misleading for aggregate outcomes.

9.5 Limitations

The paper is a quantitative model exercise rather than a causal empirical identification design. The pricing-to-market result depends on the nested CES structure, market-share-dependent markups, and the exact form of competition. The model abstracts from some features that may matter in practice, such as multiproduct firms, inventories, distribution networks beyond the stylized distribution-cost extension, currency invoicing, and nominal rigidities. The authors explicitly note that a more general theory of pricing-to-market remains an important topic for future research.

9.6 Takeaway

The enduring takeaway is that international price deviations can be generated by real-side market structure. Trade costs segment markets, variable markups make prices respond less than costs, and heterogeneous large firms translate micro pricing decisions into aggregate deviations from PPP.